



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 3: Analyze data in a data warehouse

- **Create a workspace**
- **Create a data warehouse**
- **Create tables and insert data**
- **Query data warehouse tables**
- **Create a view**
- **Create a visual query**
- **Define a data model**



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Create a workspace

The screenshot shows the 'Create a workspace' dialog in the Microsoft Fabric application. The dialog is open over the main Fabric interface, which includes a sidebar with navigation options like Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, and Workloads. The main area displays 'Create a workspace with predefined task flow' options and a 'Quick access' section with recent workspaces.

Create a workspace

Name *
Analyze_data_WH
This name is available

Description
Describe this workspace

Domain
Assign to a domain (optional)

Learn more about workspace settings

Workspace image
Upload
Reset

Advanced

Contact list *
rsidvika (Owner)

Workspace type
What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types
Power BI workspaces only support the use of Power BI capabilities.

Apply **Cancel**

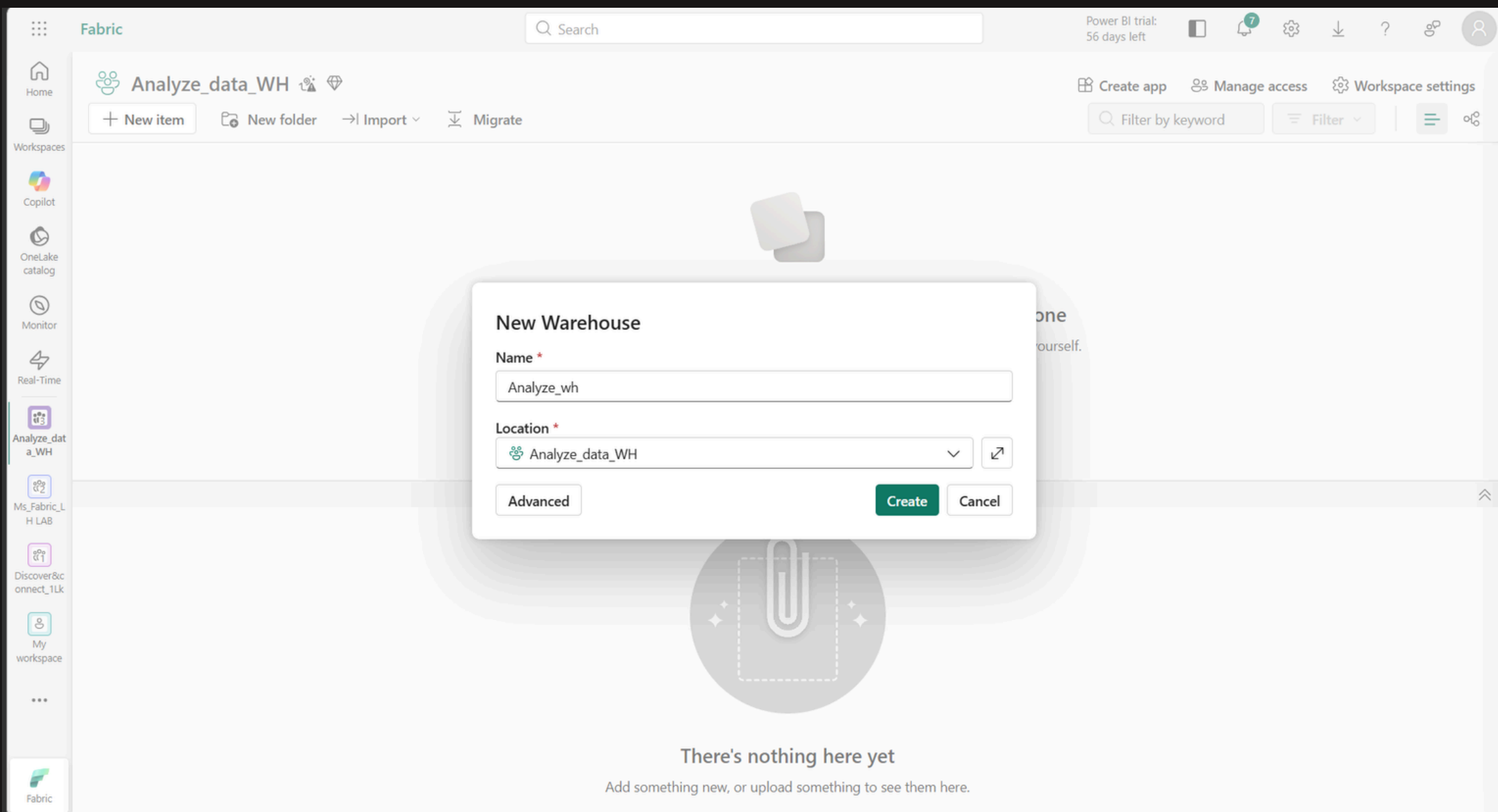
WS name: Analyze_data_Ws

WS Type: Trial



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➤ Create a data warehouse



WH name: Analyze_wh



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➤ Create tables and insert data

Analyze_wh

Search

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Explorer

+ Warehouses

Analyze_wh

Schemas

dbo

Tables

DimCustomer

DimDate

DimProduct

FactSalesOrder

Views

Functions

Stored Procedures

INFORMATION_SCHE...

queryinsights

sys

Security

Queries

Copy SQL connection string Succeeded (5 sec 320 ms)

Columns: 10 Rows: 407

Data preview - DimCustomer

Showing 1000 rows

	123 CustomerK...	ABC Customer...	ABC Title	ABC FirstName	ABC LastName	ABC AddressLin...	ABC City	ABC StateProvi...	ABC CountryRe...	ABC PostalCode
1	29546	christopher1@...	Mr.	Christopher	Beck	93-2501, Blackf...	London	England	United Kingdom	SE1 8HL
2	29587	john12@adven...	Mr.	John	Brooks	99-3 Forest Wo...	London	England	United Kingdom	E17 6JF
3	29612	richard2@adve...	Mr.	Richard	Byham	Knaves Beech I...	High Wycombe	England	United Kingdom	HP10 9QY
4	29615	mari0@advent...	Ms.	Mari	Caldwell	2524 Barlby Ro...	London	England	United Kingdom	W10 6BL
5	29618	gustavo1@adv...	Sr.	Gustavo	Camargo	Ellesfield Ave	Bracknell	England	United Kingdom	RG12 8TB
6	29644	brigid0@adven...	Ms.	Brigid	Cavendish	Banbury	Oxon	England	United Kingdom	OX16 8RS
7	29713	dick1@adventu...	Mr.	Dick	Dievendorff	Bradford	W. York	England	United Kingdom	BD1 4SJ
8	29736	terry1@advent...	Mr.	Terry	Eminhizer	Warrington Ldc...	Woolston	England	United Kingdom	WA1 4SY
9	29741	janeth1@adve...	Sra.	Janeth	Esteves	6 The Farm, Lib...	Maidenhead	England	United Kingdom	SL67RJ
10	29776	barbara4@adv...	Ms.	Barbara	German	Science Park So...	Warrington	England	United Kingdom	WA3 7BH
11	29778	leo0@adventur...	Mr.	Leo	Giakoumakis	Elmhurst Road, ...	Reading	England	United Kingdom	RG7 5H7
12	29796	jon1@adventur...	Mr.	Jon	Grande	Galashiels	Liverpool	England	United Kingdom	L4 4HB
13	29851	jean2@advent...	Ms.	Jean	Holloway	Unit 99a Batter...	Birmingham	England	United Kingdom	B29 6SL
14	29857	janice1@adven...	Ms.	Janice	Hows	399 Clearing Gr...	London	England	United Kingdom	EC1R 0DU
15	29860	arlene0@adven...	Ms.	Arlene	Huff	2574 Milton Park	Oxford	England	United Kingdom	OX14 4SE
16	29868	denean0@adve...	Ms.	Denean	Ison	586 Fulham Ro...	London	England	United Kingdom	SW6 SBY
17	29874	david17@adve...	Mr.	David	Jaffe	Wade Road	Basingstoke Ha...	England	United Kingdom	RG24 8PL
18	29904	mary8@advent...	Ms.	Mary	Kesslep	Wharfdale Road	Berkshire	England	United Kingdom	RG11 5TP
19	29908	joe1@adventur...	Mr.	Joe	Kim	308-3250 Casti...	York	England	United Kingdom	YO24 1GF
20	29922	pamala0@adve...	Ms.	Pamala	Kotc	Garamonde Dri...	Milton Keynes	England	United Kingdom	MK8 8ZD
21	29926	deepak0@adve...	Mr.	Deepak	Kumar	255 Irving Street	London	England	United Kingdom	C2H 7AU
22	29932	rebecca2@adv...	Ms.	Rebecca	Laszlo	Phoenix Way, C...	Gloucestershire	England	United Kingdom	GL7 1RY

Tables

DimCustomer

DimDate

DimProduct

FactSalesOrder



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► Query data warehouse tables

The screenshot shows the Microsoft Fabric SQL query editor interface. The top bar includes a search bar, a 'Power BI trial: 55 days left' notification, and a 'Share' button. The left sidebar contains navigation icons for Home, Copilot, Create, Browse, Apps, Scorecards, OneLake catalog, Learn, Workspaces, and Analyze data. The main area displays a query in 'SQL query 3' with the following SQL code:

```
1 Select d.[Year] AS CalendarYear,  
2 d.[Month] AS MonthofYear,  
3 d.MonthName AS MonthName,  
4 SUM(so.SalesTotal) AS SalesRevenue  
5 FROM FactSalesOrder AS so  
6 Join DimDate AS d ON so.SalesOrderDateKey=d.DateKey  
7 Group by d.[Year], d.[Month], d.MonthName  
8 ORDER BY CalendarYear, MonthofYear;
```

Below the query editor, the 'Results' tab is active, showing a table with 4 columns: CalendarYear, MonthofYear, MonthName, and SalesRevenue. The table contains 24 rows of data for the year 2021.

	123 CalendarYear	123 MonthofYear	ABC MonthName	e ^x SalesRevenue
1	2021	1	January	687029
2	2021	2	February	836937
3	2021	3	March	869863
4	2021	4	April	916273
5	2021	5	May	961241
6	2021	6	June	659677
7	2021	7	July	572189
8	2021	8	August	794832
9	2021	9	September	929086
10	2021	10	October	614116
11	2021	11	November	890905
12	2021	12	December	820810
13	2021	1	January	746060

The bottom status bar indicates 'AutoSave: On (Saved)', 'Succeeded (1 sec 819 ms)', and 'Copilot completions: Off'. The bottom right corner shows 'Columns: 4 Rows: 24'.

Query

```
SELECT d.[Year] AS CalendarYear,  
       d.[Month] AS MonthOfYear,  
       d.MonthName AS MonthName,  
       SUM(so.SalesTotal) AS SalesRevenue  
FROM FactSalesOrder AS so
```

```
JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey  
GROUP BY d.[Year], d.[Month], d.MonthName  
ORDER BY CalendarYear, MonthOfYear;
```




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Create a View

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Search

Analyze data with Share

Explorer

DimCustomer DimDate DimProduct FactSalesOrder SQL query 3 SQL query 4

Run Save as Save as view Create rule Explain query Fix query errors

```
1 --
2
3
4 CREATE VIEW vSalesBYRegion
5 AS
6 SELECT d.[Year] AS CalendarYear,
7        d.[Month] AS MonthofYear,
8        d.MonthName AS MonthName,
9        c.CountryRegion AS SalesRegion,
10       SUM(so.SalesTotal) AS SalesRevenue
11
12 From FactSalesOrder AS so
13 Join DimDate AS d ON so.SalesOrderDateKey=d.DateKey
14 JOIN DimCustomer AS c ON so.CustomerKey = c.CustomerKey
15 Group BY d.[Year], d.[Month], d.MonthName, c.CountryRegion;
```

Messages Copy messages

3:38:00 PM Started executing on line 1
3:38:04 PM Query execution time: 00:00:00.010 | Time to process and render results: 00:00:03.444 | Total duration: 00:00:03.454

Power BI

AutoSave: On (Saved) Succeeded (3 sec 454 ms) Copilot completions: Off

DimCustomer DimDate DimProduct FactSalesOrder SQL query 3 SQL query 4

Run Save as Save as view Create rule Explain query Fix query errors

```
1
2
3 SELECT CalendarYear, MonthName, SalesRegion, SalesRevenue
4 From vSalesBYRegion
5 ORDER BY CalendarYear,MonthofYear, SalesRegion;
```

Messages Results

	123 CalendarYear	ABC MonthName	ABC SalesRegion	e* SalesRevenue
1	2021	January	Canada	78554
2	2021	January	United Kingdom	31025
3	2021	January	United States	226684
4	2021	February	Canada	51482
5	2021	February	United Kingdom	17950
6	2021	February	United States	341371
7	2021	March	Canada	126748
8	2021	March	United Kingdom	18652
9	2021	March	United States	205325





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Create a visual query

Analyze_wh

Search

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Home Management Help

Analyze data with Share

DimCustomer DimDate DimProduct FactSalesOrder SQL query 3 SQL query 4 Visual query 1

Manage columns Reduce rows Sort Transform Combine Save as view View SQL

DimProduct FactSalesOrder

Source Database Table Merged queries Expanded DimPr...

Save as table Download Excel file Visualize results

	ProductKey	ProductAltKey	ProductName	Category	ListPrice
1	1	RING1	Bicycle bell	Accessories	6
2	2	BRITE1	Front light	Accessories	15
3	3	BRITE2	Rear light	Accessories	15
4	707	HL-U509-R	Sport-100 Helmet, Red	Helmets	35
5	708	HL-U509	Sport-100 Helmet, Black	Helmets	35
6	709	SO-B909-M	Mountain Bike Socks, M	Socks	10
7	710	SO-B909-L	Mountain Bike Socks, L	Socks	10
8	711	HL-U509-B	Sport-100 Helmet, Blue	Helmets	35
9	712	CA-1098	AWC Logo Cap	Caps	9
10	713	LJ-0192-S	Long-Sleeve Logo Jersey, S	Jerseys	50
11	714	LJ-0192-M	Long-Sleeve Logo Jersey, M	Jerseys	50
12	715	LJ-0192-L	Long-Sleeve Logo Jersey, L	Jerseys	50
13	716	LJ-0192-X	Long-Sleeve Logo Jersey, XL	Jerseys	50
14	720	FR-M945-A2	HL Mountain Frame - Silver, A2	Mountain Frames	1365

Columns: 5 Rows: 99+

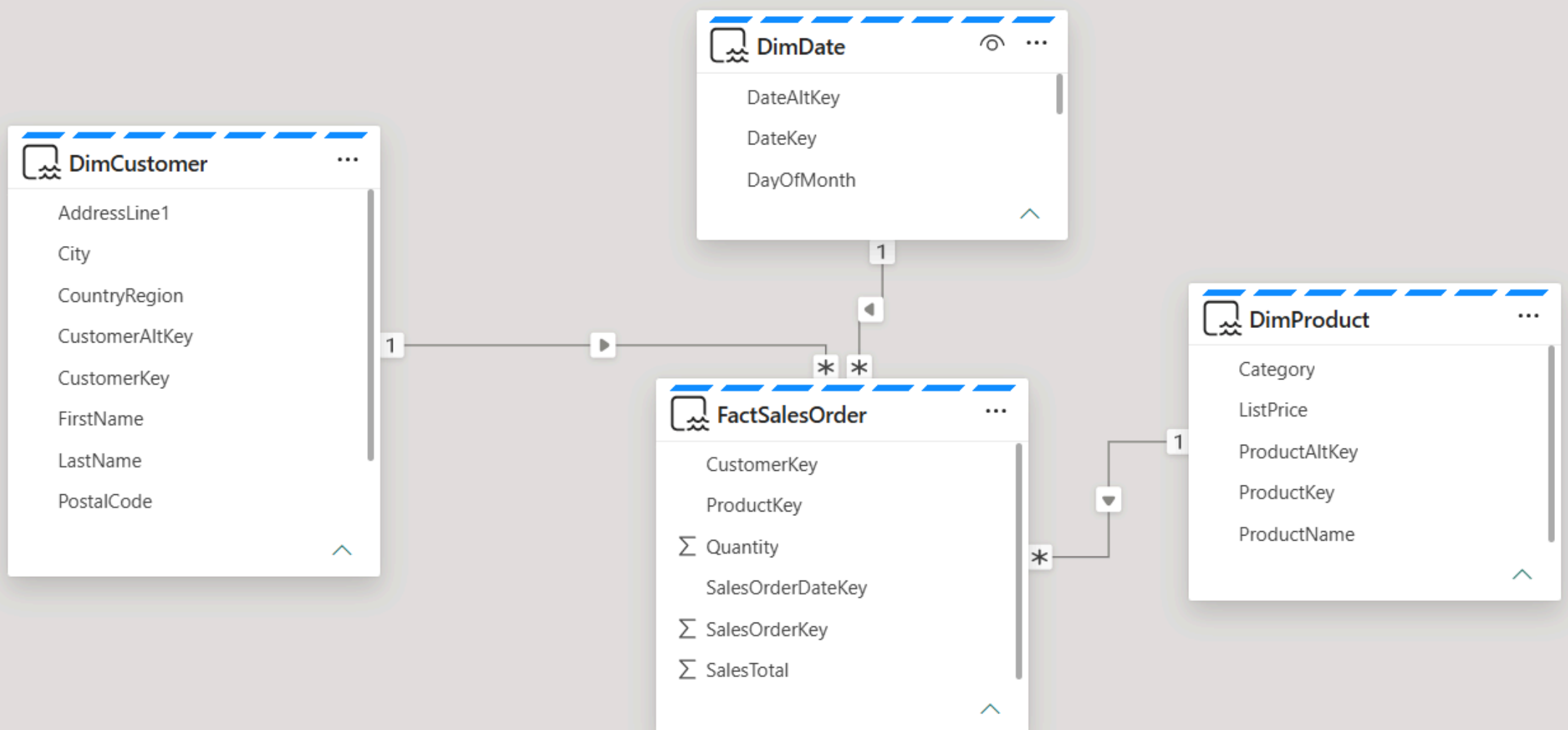
Copy SQL connection string



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Define a data model





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Reference

